PHASEFORGE
ELECTRICAL & POWER

DASHBOARD-STYLE WEBSITE IMPLEMENTATION SPECIFICATION

A practical developer brief based on your five visual references and the six approved service categories.

POWER DONE RIGHT, every single time.

Prepared for: Business owner and website developer

Currency and operating context: Ghana / GHS

Purpose of this document

Specify how every major PhaseForge website section should be designed and implemented using a consistent dashboard-like visual language: clear cards, summaries, tables, statuses, filters, step-based forms and meaningful actions. It is a build specification—not a deployed website, legal opinion or proof that current integrations work.

Important localization note

The references include UK-specific content and other example brands. Do not copy their credentials, currency, address, contact numbers, testimonials or claims. Replace them with verified PhaseForge information for Ghana.

GUIDE

Specification map

The pages below cover the public website, customer journeys, staff operations and implementation controls.

SECTION	WHAT THE DEVELOPER BUILDS
01 / Experience design — Design system & shared dashboard language	Every major section should feel like part of one professional electrical-services platform—not a collection of unrelated templates.
02 / Public website — Homepage dashboard-style experience	The homepage should quickly explain the business, build trust honestly and route visitors to services, product sales or a booking request.
03 / Services — Six service-category pages	Keep the approved six-category structure from the reference image. Each category opens a detailed page with its own service list, process, FAQs and quote/request action.
04 / Booking — Booking and quote request workspace	Turn the current booking form into a clear multi-step flow that captures useful information without promising an unconfirmed time.
05 / Shop — Electrical materials catalogue & shop	Add a proper retail catalogue without making it look like a generic marketplace. The product experience should be as organized as the dashboard references.
06 / Admin overview — Main operations dashboard	This is the control centre for the owner and authorized staff. The dashboard should use cards, tables, filters and action panels similar to the reference screenshot.
07 / Job management — Job lifecycle, dispatch & technician workspace	Every service request should move through a controlled lifecycle so customers, technicians and office staff know the current state.
08 / Customer portal — Customer account and job tracking	Give customers a simple, secure way to see their own requests, quotes, appointments, invoices and service history.
09 / Quotes & pricing — Quotation builder and approval controls	A quote should be easy to understand, versioned and explicit about assumptions, exclusions, validity and payment milestones.
10 / Invoices & payments — Financial dashboard and payment reconciliation	Financial status must be derived from invoices and verified payment records—not from a customer clicking a button.
11 / Inventory & purchasing — Stock, suppliers and job materials	The catalogue and inventory must connect retail sales to service jobs, while keeping the cost of stock private from customers.
12 / Project contracts — Projects, developers and maintenance agreements	Support larger jobs with milestones, site records, change orders and recurring maintenance—not just one-off bookings.
13 / Policies — Customer terms, safety and service policies	Policies must be readable at the point they matter, with version control and evidence of customer acknowledgement.
14 / Notifications — Email, WhatsApp and reminders	Notifications should help the customer and team coordinate, while the system remains truthful about delivery and confirmation.
15 / Reports — Management reports and business intelligence	Reports should help the owner understand service demand, cash flow, profitability, response performance and stock needs.
16 / Security — Security, roles, privacy and backups	The site handles customer addresses, photographs, business records and payments. Security must be part of implementation, not a later decoration.
17 / Technical architecture — Recommended implementation architecture	Keep the existing website where sensible, but make sure the user interface is backed by secure, maintainable services.
18 / Content & Ghana localization — Ghana-first content, contacts and operating boundaries	Keep the strong visual references, but replace UK-specific content with accurate Ghanaian information and only verified business promises.
19 / Testing — Quality assurance and launch acceptance	The website is not complete when it looks correct. It is complete when the critical journeys work safely and consistently on real devices.
20 / Delivery roadmap — Build in phases without losing control	Implement the platform in a sequence that delivers usable value early and avoids building a visually impressive but unreliable system.

SECTION	WHAT THE DEVELOPER BUILDS
21 / Reference mapping — How the supplied visual references translate to PhaseForge	Use the uploaded references as visual direction for layout and hierarchy, not as content to copy verbatim.
22 / Developer handover — Required handover package	The owner should be able to operate the business and change routine content without needing the developer for every update.

Global requirement: all visible dashboard metrics, product stock, appointment slots, payment states and customer records must be driven by real data and permission-checked backend logic. Never leave demo values in production.

01 / EXPERIENCE DESIGN

Design system & shared dashboard language

Every major section should feel like part of one professional electrical-services platform—not a collection of unrelated templates.

BRAND

PhaseForge

POWER DONE RIGHT, every single time

STYLE

Dark navy + gold

White cards, clear spacing

LAYOUT

Responsive

Desktop, tablet and mobile

LANGUAGE

Ghana-first

GHS, Ghana locations, local contact routes

Interface modules

Header and navigation

Brand logo, Home, Services, Shop, Projects, About, Contact, Book a Service. On small screens use a clear menu and a persistent booking action.

Dashboard cards

Use compact summary cards for counts, statuses, prices and next actions. Cards must show real data, not decorative fake metrics.

Typography and spacing

High contrast, readable sizes, consistent spacing, visible focus states and large touch targets.

Status language

Use plain statuses: New request, Awaiting confirmation, Scheduled, Assigned, In progress, Awaiting customer, Completed, Invoiced, Paid, Cancelled.

Expected workflow

STAGE	BEHAVIOR / RULE
Desktop	Sidebar or top navigation; main content uses KPI cards, filters, charts/tables and action panels.
Mobile	Single-column cards; tables become stacked rows; primary actions remain visible; no horizontal scrolling for essential tasks.
Accessibility	Keyboard navigation, labels for all fields, sufficient contrast, alt text and helpful validation errors.

Acceptance checklist

- ✓ Build reusable UI components instead of styling each page independently.
- ✓ Use the same spacing, button hierarchy, badges, form fields and card patterns throughout.

✓ Do not publish invented testimonials, credentials, customer counts or performance figures.

02 / PUBLIC WEBSITE

Homepage dashboard-style experience

The homepage should quickly explain the business, build trust honestly and route visitors to services, product sales or a booking request.

PRIMARY ACTION

Book a Service

Request a visit or quote

SECOND ACTION

Shop Products

Browse electrical materials

TRUST MESSAGE

POWER DONE RIGHT

Every single time

COVERAGE

Service areas

Display verified areas only

Interface modules

Hero panel

Dark navy panel with gold accents, a strong headline, one authentic electrical-work image, a short Ghana-relevant explanation and two clear actions: Book a Service and Shop Materials.

Service shortcuts

Six visual cards: Home Electrical Services; Commercial & Industrial; Solar & Backup Power; Testing & Safety; Emergency Assistance; Projects & Contracts.

Featured products

Show selected products with image, brand/specification, price or quote-only label, stock status and product details link.

How it works

Request → review → quote/approval → schedule → perform work → test/handover → invoice/payment → follow-up.

Proof and reassurance

Use genuine project photos, verifiable qualifications, clear service terms and real reviews with consent.

Contact strip

Call/WhatsApp options, email, service areas, hours and emergency availability—only where the business can deliver.

Expected workflow

STAGE	BEHAVIOR / RULE
Visitor chooses	Service booking, product enquiry, quotation, emergency contact or project/contract enquiry.
Website responds	Explain what happens next and distinguish a request received from a confirmed appointment.
Admin receives	Create a traceable request in the staff dashboard and notify authorized staff.

Acceptance checklist

✓ All primary buttons go to working pages or actions.
✓ No unsupported claims such as guaranteed same-day response, free assessment or 24/7 availability unless operationally true.
✓ Use mobile-optimized images and avoid slowing the homepage.

03 / SERVICES

Six service-category pages

Keep the approved six-category structure from the reference image. Each category opens a detailed page with its own service list, process, FAQs and quote/request action.

CATEGORIES

6

Approved top-level groups

SERVICE PAGES

Detail

Each has scoped service items

REQUEST FORM

Contextual

Preselects the chosen service

SAFETY

First

Clear limits and urgent guidance

Interface modules

Home Electrical Services

House wiring and rewiring; fault diagnosis; socket/switch/lighting work; distribution boards; repairs; inspections; maintenance.

Commercial & Industrial

Shops, offices and commercial premises; distribution and wiring; maintenance; three-phase work, control panels and industrial faults only where qualified.

Solar & Backup Power

Site assessment; solar/inverter/battery planning and installation; generator/changeover systems; backup-power troubleshooting and maintenance.

Testing & Safety

Electrical condition assessments, protective-device checks, earthing/bonding checks, testing and written findings. Formal certificates only by authorized persons.

Emergency Assistance

Urgent electrical-fault enquiries, safe isolation guidance and dispatch subject to availability. Tell customers to keep away from sparking/live equipment.

Projects & Contracts

New builds, renovations, multi-unit properties, bulk materials supply, contractor coordination and scheduled maintenance agreements.

Expected workflow

STAGE	BEHAVIOR / RULE
Category detail	Describe scope, common symptoms/use cases, exclusions, preparation, typical process and what affects price.
CTA	Request a quote / Book assessment / Ask about a contract. Pass category and service into booking form.
Qualification gate	Admin can mark a service as offered, temporarily unavailable or referral-only; hide or label unsupported work.

Acceptance checklist

✓ No copied UK-specific credentials, currencies or claims; localize all content for Ghana.

✓ Keep niche items only if actually offered and staff have the right competence, tools and approvals.
✓ Every service page includes safety notes and a clear next action.

04 / BOOKING

Booking and quote request workspace

Turn the current booking form into a clear multi-step flow that captures useful information without promising an unconfirmed time.

STEP 1

Service

Category and job type

STEP 2

Site details

Location, access, urgency

STEP 3

Review

Consent and submit

REFERENCE

Generated

Unique request number

Interface modules

Step 1 — What is needed?

Choose service category and specific job; describe fault or desired outcome; choose residential/commercial/industrial/project; indicate whether materials are needed.

Step 2 — Where and when?

Customer name, phone/WhatsApp, email if available, Ghana region/town, full site directions, landmark, preferred date/time window and urgency.

Step 3 — Supporting evidence

Optional photos of safe-to-photograph areas or equipment labels. Warn customers not to remove covers or touch live equipment to take photos.

Step 4 — Review and submit

Show a summary, privacy notice, contact preference, consent checkbox and clear statement that submission is a request, not yet a confirmed appointment.

Confirmation

Generate request reference; show next steps and contact details. Send only after backend has successfully saved the request.

Admin triage

New request enters queue; authorized staff assess urgency, coverage, capability, missing details and whether a site visit or quote is needed.

Expected workflow

STAGE	BEHAVIOR / RULE
Emergency	Show call/WhatsApp options and safety instructions; do not imply the online form is monitored instantly unless it is.
Availability	Only offer time slots that are genuinely available and protected against double-booking.
Change/cancel	Customer can request a change; staff confirms revised booking and records the change history.

Acceptance checklist

✓ Server validates every field and upload; files have size/type limits and private access controls.

✓ Booking reference is unique; failed saves show a retry message and do not falsely claim success.
✓ Staff notification and customer confirmation are logged with delivery status.

05 / SHOP

Electrical materials catalogue & shop

Add a proper retail catalogue without making it look like a generic marketplace. The product experience should be as organized as the dashboard references.

CATALOGUE

Search + filters

Category, brand, use

CURRENCY

GHS

Tax/delivery rules configured

STOCK

Live status

In stock, low, on request

ORDER PATH

Cart or enquiry

Depends on fulfilment setup

Interface modules

Product categories

Cables/wires; breakers and protection; switches/sockets/plugs; conduits/fittings; LED lighting; tools/testing equipment; solar/inverters/batteries; generators; accessories.

Product cards

Image, product name, manufacturer/brand, key rating/specification, unit of sale, price or quote-only, stock status and View Details.

Product detail

SKU, full specifications, compatibility notes, warranty/returns information, available quantities, delivery/pickup options and enquiry CTA.

Cart and order request

Quantity, unit, subtotal, delivery/pickup details, customer contact and clear order status. Do not treat a stock enquiry as a paid order.

On-demand equipment

For high-value solar, inverter, battery and generator equipment, label “Request a quote / availability” until price and supplier stock are confirmed.

Stock control

SKU, category, purchase cost, selling price, quantity, reorder threshold, supplier, batch/serial if relevant and movement history.

Expected workflow

STAGE	BEHAVIOR / RULE
Retail sale	Order request → stock check → quote/confirmation → payment method → pick/pack → delivery/pickup → receipt.
Stock change	Record purchase, sale, return, adjustment or damaged item with user, timestamp and reason.
Product unavailable	Offer an enquiry or restock notification; never show unavailable stock as guaranteed available.

Acceptance checklist

✓ Prices are managed by authorized staff and display effective dates or last-updated details internally.
✓ Do not expose supplier purchase cost to public users.
✓ Product images and technical specifications must match the actual product.

06 / ADMIN OVERVIEW

Main operations dashboard

This is the control centre for the owner and authorized staff. The dashboard should use cards, tables, filters and action panels similar to the reference screenshot.

NEW REQUESTS

Live count

Awaiting triage

SCHEDULED JOBS

Live count

Upcoming visits

JOBS IN PROGRESS

Live count

Active work

OUTSTANDING INVOICES

Live value

Amounts due

Interface modules

Top KPI row

New requests, jobs scheduled today, jobs in progress, overdue tasks, unpaid invoices, low-stock products. Every number is computed from database records.

Work queue

A prioritized table with request ID, customer, location, service, urgency, status, assigned technician and next action.

Calendar / dispatch

Day/week view for confirmed appointments, technician workload, travel buffers and unavailable periods.

Finance panel

Quotes awaiting approval, invoices due, payments received, overdue balances and refunds requiring review.

Inventory alerts

Low-stock list, stockouts, pending supplier orders, items reserved for jobs and stock discrepancies.

Recent activity

Auditable feed of job status changes, quote approvals, stock movements, payment updates and admin actions.

Expected workflow

STAGE	BEHAVIOR / RULE
Role-based views	Owner sees all authorized business metrics; dispatcher sees jobs; store staff sees inventory; finance sees invoices/payments; technicians see assigned work.
Filters	Date range, service category, status, town/region, technician, customer and payment status.
Quick actions	Create job, assign technician, draft quote, create invoice, add product, record stock receipt—each action checks permissions.

Acceptance checklist

✓ All KPI cards link to the records behind the number.

✓ Empty, loading, error and permission-denied states are designed.
✓ Dashboard metrics never use sample/demo data in production.

07 / JOB MANAGEMENT

Job lifecycle, dispatch & technician workspace

Every service request should move through a controlled lifecycle so customers, technicians and office staff know the current state.

INTAKE

New

Request received

PLANNING

Scheduled

Time confirmed

EXECUTION

In progress

Work underway

CLOSEOUT

Completed

Test, handover, invoice

Interface modules

Job record

Unique job ID, linked customer, site address, service category, description, urgency, photos, risk notes, assigned staff, dates, materials, quote and invoice.

Assignment

Assign qualified technician/team; confirm acceptance; prevent assignment to unavailable staff; record reassignment reason.

On-site checklist

Arrival, customer/site confirmation, safe isolation where required, scope confirmation, pre-existing conditions, materials used and work performed.

Change control

If the scope or price changes, document reason, revised quote, customer approval and schedule impact before proceeding where practicable.

Testing and handover

Record checks appropriate to the job, test results, unresolved defects, customer explanation, photos where consented and completion acknowledgement.

Follow-up

Warranty/return visit eligibility, customer feedback request, maintenance reminder and any outstanding action.

Expected workflow

STAGE	BEHAVIOR / RULE
Request → triage	Confirm area, urgency, capability, customer details and whether a site visit is needed.
Quote → approval	Save quote version and approval evidence; do not silently alter an accepted quote.
Schedule → work	Confirm slot, assign technician, send appropriate updates and record arrival/progress.
Complete → invoice	Complete required checklist, record materials and labour, issue invoice and set payment status.

Acceptance checklist

✓ Status transitions are controlled and logged with user/time.
✓ Technicians can update only permitted assigned jobs.
✓ A job cannot be marked complete if required closeout fields are missing, unless an authorized override is recorded.

08 / CUSTOMER PORTAL

Customer account and job tracking

Give customers a simple, secure way to see their own requests, quotes, appointments, invoices and service history.

REQUESTS

My requests

Status and reference

QUOTES

Review

Approve or ask questions

APPOINTMENTS

Upcoming

Time and site details

INVOICES

Balance

Payment status and receipt

Interface modules

Customer home

Upcoming appointment, open request, quote awaiting response, outstanding invoice and latest job update.

Request history

Customer sees only their own requests, attachments, job notes marked customer-visible and agreed documents.

Quote approval

Scope, exclusions, line items, validity date, deposit/milestone terms, approve/decline/request clarification and record timestamp.

Messages and updates

Use approved email/WhatsApp channels; show sent/failed state where possible and keep staff-only notes private.

Documents

Download quotes, invoices, receipts, job completion summaries and maintenance recommendations securely.

Profile and privacy

Edit contact details, notification preferences and request account/data assistance through a documented process.

Expected workflow

STAGE	BEHAVIOR / RULE
Identity	Use verified sign-in or a secure one-time link; do not rely on guessable job IDs alone.
Access	Enforce ownership on the server for every record, file and document request.
Notifications	Send minimal necessary information; do not put sensitive job details in public links.

Acceptance checklist

- ✓ Customer A cannot view Customer B's jobs, files, addresses or invoices.
- ✓ Account recovery and phone/email changes require appropriate verification.
- ✓ Customers can continue to contact the business without creating an account if the owner chooses to support guest requests.

09 / QUOTES & PRICING

Quotation builder and approval controls

A quote should be easy to understand, versioned and explicit about assumptions, exclusions, validity and payment milestones.

DRAFTS

Pending

Internal review

SENT QUOTES

Awaiting

Customer response

APPROVED

Authorized

Scope may be scheduled

EXPIRED

Review

Reprice before work

Interface modules

Quote header

Quote number, customer, site, issue date, validity date, prepared by, related request/job and contact details.

Line items

Labour, materials, quantities, unit, unit price, tax treatment if applicable, discount if authorized, subtotal and total.

Scope & exclusions

Work included, work not included, assumptions, access conditions, making-good/finishing, disposal, third-party fees and unknown faults.

Payment terms

Deposit if any, milestone dates, balance due, accepted payment channels, late-payment handling and cancellation implications.

Approval evidence

Customer approval method, version, date/time and approved amount. Changes create a new version or variation.

Pricing rules

Fixed prices for standardized tasks only when scope is known; diagnostic/call-out fees disclosed before dispatch; custom quotes for complex work.

Expected workflow

STAGE	BEHAVIOR / RULE
Create	Draft quote from request/job and product/labour catalogue.
Review	Authorized manager checks pricing, scope, margin and any required site inspection.
Send	Customer receives readable PDF or portal quote with expiry and approval actions.
Revise	Keep history; record reason and customer acknowledgement of material changes.

Acceptance checklist

✓ No quote is labelled free unless the business intends to provide it free in that situation.
✓ Estimate versus fixed quote is clearly distinguished.
✓ Do not add unapproved work or costs without customer consent except necessary emergency safety action handled under agreed terms.

10 / INVOICES & PAYMENTS

Financial dashboard and payment reconciliation

Financial status must be derived from invoices and verified payment records—not from a customer clicking a button.

INVOICED

Live total

For selected period

RECEIVED

Verified

Reconciled payments

OUTSTANDING

Due balance

By due date

REFUNDS

Review queue

Authorized approval

Interface modules

Invoice builder

Invoice number, customer, job, quote link, itemized labour/materials, issue/due dates, totals, payment instructions and terms.

Payment channels

Configure only providers/channels supported for the business in Ghana; present Mobile Money, bank transfer or cash options only when actually accepted.

Server-side verification

Payment status changes only after trusted provider confirmation or documented staff reconciliation. Never trust a browser redirect alone.

Receipts

Issue receipt after verified payment; include amount, currency, reference, date and remaining balance if partial.

Refund and reversal

Role-restricted request, reason, original payment reference, approval and final reconciliation.

Financial audit

Record edits, voids, write-offs, discounts, refunds and manual payment entries with user/time/reason.

Expected workflow

STAGE	BEHAVIOR / RULE
Invoice → payment	Issue invoice; customer chooses available method; payment is verified; ledger is updated; receipt is issued.
Partial payment	Record amount against invoice and calculate remaining balance accurately.
Failed/pending	Show pending or failed without marking paid; offer instructions for support and safe retry.

Acceptance checklist

- ✓ Use GHS display and agreed tax treatment reviewed by a Ghanaian accountant.
- ✓ No card/MoMo secrets or payment credentials stored in application logs.

✓ Daily reconciliation report can compare provider records, invoices and bank/cash entries.

11 / INVENTORY & PURCHASING

Stock, suppliers and job materials

The catalogue and inventory must connect retail sales to service jobs, while keeping the cost of stock private from customers.

ON HAND

Quantity

Current counted stock

LOW STOCK

Alerts

Reorder threshold

RESERVED

For jobs

Not available for sale

PURCHASING

Open orders

Supplier deliveries

Interface modules

Product master

SKU, description, unit, brand, technical ratings, selling price, cost, preferred supplier and reorder level.

Goods received

Supplier, supplier invoice, quantities, cost, condition, batch/serial details and receiving staff.

Stock movement

Every sale, reservation, job issue, return, damage, correction and transfer has a timestamp and reason.

Job material issue

Reserve stock for an approved job; record actual usage; return unused materials; attach cost to job profitability.

Supplier directory

Business name, contact, products, lead times, quote history and notes; restrict sensitive commercial information.

Stocktake

Count physical quantities, compare system values, approve adjustments and report recurring discrepancies.

Expected workflow

STAGE	BEHAVIOR / RULE
Buy	Draft purchase request → approve → order → receive and inspect → stock ledger update.
Use	Reserve for job or sale → issue/dispatch → record actual quantity → reconcile leftovers/returns.
Reorder	Use thresholds and recent usage as prompts; owner approves purchase based on cash flow and demand.

Acceptance checklist

- ✓ Prevent negative stock unless an authorized controlled exception is required.
- ✓ Keep stock ledger history; do not simply overwrite quantity without a reason.
- ✓ High-value equipment can be quote-only or supplier-confirmed rather than held as stock.

12 / PROJECT CONTRACTS

Projects, developers and maintenance agreements

Support larger jobs with milestones, site records, change orders and recurring maintenance—not just one-off bookings.

PROJECTS

Active

Milestones and owners

CONTRACTS

Current

Scope and term

VARIATIONS

Approval

Changes to scope/cost

MAINTENANCE

Due

Upcoming visits

Interface modules

Project overview

Project ID, client, property/site, drawings/documents, contacts, target dates, scope, budget, responsible manager and current risks.

Milestone plan

Survey, design/quotation, procurement, first-fix, second-fix, testing, handover and payment milestones as applicable.

Contract terms

Scope, deliverables, exclusions, responsibilities, access, materials, milestones, change procedure, delay handling, warranty and termination.

Site diary

Daily work summary, attendance, progress, blockers, delivery records, photos with permission and client instructions.

Variation orders

Change description, price/time effect, approval status and linked revised drawings/quote.

Maintenance plans

Asset/site, schedule, checklist, service report, defects, next visit and renewal reminder.

Expected workflow

STAGE	BEHAVIOR / RULE
Enquiry	Capture project size, location, drawings, target dates, procurement needs and decision maker.
Assessment	Arrange site visit, assess scope and capability, prepare proposal and realistic schedule.
Delivery	Manage milestones, quality checks, materials, customer approvals and payment milestones.
Handover	Provide agreed test records, manuals, warranties, as-built information where applicable and maintenance plan.

Acceptance checklist

✓ Do not imply engineering design, certification or specialist capability unless qualified and authorized.
✓ Contracts should be reviewed for Ghanaian legal and tax context before use.
✓ Every variation is documented; avoid informal chat-only scope changes for significant cost changes.

13 / POLICIES

Customer terms, safety and service policies

Policies must be readable at the point they matter, with version control and evidence of customer acknowledgement.

SERVICE TERMS

Versioned

Scope, access, changes

PRIVACY

Clear

Data and photo use

WARRANTY

Defined

Workmanship vs products

COMPLAINTS

Tracked

Owner and resolution date

Interface modules

Booking terms

Request versus confirmation, appointment windows, access, rescheduling, cancellation and no-show handling.

Quote and payment terms

Estimate/fixed quote, validity, deposits, variation approval, invoice timing, payment channels, overdue balances and refunds.

Safety and access

Customer responsibilities, hazards, safe access, pets/occupants, power interruption and stop-work rights when unsafe.

Workmanship and product warranty

State actual warranty period and exclusions after legal/business review; distinguish labour workmanship from manufacturer product warranty.

Privacy and media

Purpose for collecting contact details/photos, access, retention, sharing, security and consent for marketing/public project images.

Complaints

Acknowledgement, investigation, response target, escalation route, corrective action and record retention.

Expected workflow

STAGE	BEHAVIOR / RULE
Before submit	Link short plain-language terms and privacy notice; obtain only necessary consent.
Before work	Confirm scope, price, access, safety concerns and any required interruption.
At closeout	Provide job summary, payment documents, warranty terms and complaint contact.
Policy updates	Store version and acknowledgement date; notify customers where changes materially affect ongoing agreements.

Acceptance checklist

✓ Have a Ghana-qualified lawyer review the final terms, consumer obligations, liability clauses and dispute process.

✓ Do not present the generated policy as legal advice or as automatically compliant.
✓ Never claim warranty or insurance coverage that has not been confirmed.

14 / NOTIFICATIONS

Email, WhatsApp and reminders

Notifications should help the customer and team coordinate, while the system remains truthful about delivery and confirmation.

CUSTOMER

Updates

Received, scheduled, changes

STAFF

Alerts

New request, urgent item

DELIVERY

Logged

Sent, failed, retry

CONSENT

Respected

Preference-aware messages

Interface modules

Request received

Confirmation reference and what happens next; clearly say appointment is not confirmed yet.

Appointment confirmation

Confirmed date/time window, site, contact, scope and rescheduling route.

Job progress

Assignment/arrival/progress/completion messages according to workflow and customer preference.

Quote/invoice

Secure link or attachment, due date, approved amount and support contact; avoid sensitive public links.

Internal alerts

New urgent request, overdue job, low stock, failed notification, quote expiry and payment reconciliation exception.

Retry and audit

Record provider response, retry safely, avoid duplicate spam and allow staff to see failures.

Expected workflow

STAGE	BEHAVIOR / RULE
Trigger	A valid status change or event creates a notification job.
Send	Use approved provider and template; protect secrets server-side.
Track	Store provider message ID and delivery status where available.
Fallback	If a notification fails, show it to staff; do not assume the customer received it.

Acceptance checklist

✓ Do not claim instant WhatsApp alerts unless integrated and tested.

✓ Provide an operational fallback for urgent requests.
✓ Respect applicable Ghanaian data protection and electronic communications requirements.

15 / REPORTS

Management reports and business intelligence

Reports should help the owner understand service demand, cash flow, profitability, response performance and stock needs.

OPERATIONS

Jobs

Volume, completion, delays

FINANCE

Cash flow

Receivables and margins

SALES

Products

Top sellers and stock turns

SERVICE QUALITY

Feedback

Complaints and callbacks

Interface modules

Job performance

Requests by category/location, conversion to booked jobs, completion time, cancellations, repeat visits and backlog.

Financial reports

Invoices, receipts, balances, deposits, refunds, expenses if tracked, gross margin and payment channel reconciliation.

Inventory reports

Stock valuation, low stock, stockouts, ageing inventory, movement, supplier lead time and job materials usage.

Customer reports

Repeat customers, service history, feedback, complaint resolution and maintenance renewals—limited to authorized users.

Technician/workload

Assignments, utilization, on-time arrival, job duration and rework indicators, interpreted fairly and with context.

Export

CSV/PDF export with date filters and permission checks; mark report generation time and data period.

Expected workflow

STAGE	BEHAVIOR / RULE
Filter	Select date range, category, location, staff or status.
Calculate	Use consistent definitions and authoritative records.
Review	Explain exclusions and missing data; avoid presenting incomplete periods as final.
Act	Convert insights into restocking, training, scheduling and service improvements.

Acceptance checklist

- ✓ Never expose personal customer details in public or broadly shared reports.
- ✓ Metric definitions are documented, e.g. completed job and paid invoice.

✓ Provide export permissions and log sensitive bulk exports.

16 / SECURITY

Security, roles, privacy and backups

The site handles customer addresses, photographs, business records and payments. Security must be part of implementation, not a later decoration.

ACCESS

Least privilege

Role-controlled features

FILES

Private

Authorized access only

AUDIT

Traceable

Sensitive changes logged

RECOVERY

Tested

Backups and restore plan

Interface modules

Authentication

Secure sign-in, strong password/session handling, rate limits and safe account recovery.

Authorization

Server-side checks for every job, invoice, file, staff action and admin endpoint; hiding a button is not security.

Uploads

Allow-list formats, size limits, safe storage, malware scanning where available, randomized names and no executable public uploads.

Secrets and payments

Keep API keys and provider secrets on the server; never put them in browser code or logs.

Backups

Automated database backups, documented retention, restore tests and controlled access.

Audit and incident response

Log role changes, quote/invoice edits, payment overrides, stock adjustments and suspicious access; define incident owner and response steps.

Expected workflow

STAGE	BEHAVIOR / RULE
Prevent	Use secure defaults, dependency updates, environment separation and least privilege.
Detect	Monitor errors, failed logins, payment mismatches and unusual administrative actions.
Recover	Restore from tested backup, rotate exposed secrets, document impact and communicate appropriately.

Acceptance checklist

✓ Test for cross-customer data exposure and unauthorized role escalation.

✓ Do not store full card data or passwords in plain text.
✓ Review Ghana Data Protection Act requirements and applicable obligations with appropriate professionals.

17 / TECHNICAL ARCHITECTURE

Recommended implementation architecture

Keep the existing website where sensible, but make sure the user interface is backed by secure, maintainable services.

FRONTEND

Existing site

Audit before replacing

DATABASE

Managed SQL

Relational records

FILES

Private storage

Photos and documents

INTEGRATIONS

Server-side

Payments and messaging

Interface modules

Frontend

Responsive component library for public pages, customer portal and admin workspace; shared form, table, badge, modal and card components.

Backend/API

Validate input, enforce permissions, calculate totals server-side, handle workflow transitions and emit notifications.

Database

Relational schema for customers, requests, jobs, staff, quotes, quote lines, products, stock movements, invoices, payments, contracts, policy acknowledgements and audit events.

Storage

Private bucket for photos, quotes and invoices; time-limited access links and retention rules.

Integrations

Choose Ghana-supported payment and messaging providers after checking availability, fees, webhook reliability and business account requirements.

Deployment

Separate development/staging/production, managed secrets, domain/TLS, migrations, monitoring and rollback plan.

Expected workflow

STAGE	BEHAVIOR / RULE
Audit	Developer reviews source code, current form handlers, deployment and integrations before selecting what to retain.
Build	Implement database and permission model before adding sensitive dashboard features.
Test	Use staging with fake customers/products/payments; test end-to-end before production.
Release	Back up, deploy, smoke-test, monitor errors and keep a rollback route.

Acceptance checklist

✓ Do not assume the visible form currently sends notifications or saves records; verify actual backend behavior.
✓ Use database constraints and transactions for booking, stock and payment updates.
✓ Document ownership, maintenance, environment variables and recovery steps.

18 / CONTENT & GHANA LOCALIZATION

Ghana-first content, contacts and operating boundaries

Keep the strong visual references, but replace UK-specific content with accurate Ghanaian information and only verified business promises.

CURRENCY

GHS

All prices and totals

CONTACTS

Verified

Owner-approved numbers

COVERAGE

Realistic

Areas actually served

COMPLIANCE

Reviewed

Local requirements

Interface modules

Business identity

PhaseForge Electrical & Power; slogan: “POWER DONE RIGHT, every single time.” Use approved logo, real business contact details and consistent name spelling.

Contact channels

Previously supplied details include WhatsApp 0552813116, call/WhatsApp 0245909523, emergency 0247227789 and info@phaseforge-electrical.com. Confirm ownership, availability and preferred display before launch.

Service area

Previously discussed areas include Kasoa, Tema, Kpone, East Legon and Greater Accra, with enquiries from other regions. Show travel/call-out conditions and do not promise coverage without dispatch capacity.

Hours and emergencies

Publish accurate business hours. Label emergency support as “call to check availability” unless 24/7 response is genuinely staffed.

Credentials and claims

Display only verified qualifications, licenses, insurance, years of experience, reviews, guarantees and project counts.

Language and units

GHS, Ghana town/region fields, Ghana phone formatting, local landmarks and locally relevant product specifications.

Expected workflow

STAGE	BEHAVIOR / RULE
Owner verifies	Contacts, hours, areas, services, credentials, fees, warranty and payment options.
Developer configures	One central settings area controls phone numbers, hours, service regions, currencies and policy version.
Admin maintains	Authorized staff can update products and availability without editing code.

Acceptance checklist

✓ Remove NICEIC, UK address, £ pricing and unrelated UK claims from reference inspiration.
✓ Confirm legal/business identity and local compliance statements before publication.
✓ Use genuine Ghanaian project photos or licensed images; label illustrative imagery when appropriate.

19 / TESTING

Quality assurance and launch acceptance

The website is not complete when it looks correct. It is complete when the critical journeys work safely and consistently on real devices.

FORMS

Tested

Save, errors and confirmation

PERMISSIONS

Tested

Role and customer isolation

PAYMENTS

Reconciled

Success, fail and pending

MOBILE

Tested

iOS/Android/browser widths

Interface modules

Public pages

Check links, spelling, responsive layouts, image loading, service category navigation, product search and contact links.

Booking end-to-end

Submit valid and invalid requests, upload allowed/disallowed files, simulate failed saves, confirm reference creation and staff triage.

Admin permissions

Test each role; verify no unauthorized access by direct URL or API request; verify customer records are isolated.

Shop/inventory

Test purchase, reservation, sale, return, low-stock alert, failed order and concurrent stock changes.

Finance

Test quote versions, invoice totals, partial payments, duplicate webhook, pending payment, refund and manual reconciliation.

Recovery and performance

Test backup restore, slow mobile network, large image, unavailable integration and deployment rollback.

Expected workflow

STAGE	BEHAVIOR / RULE
Staging sign-off	Owner and developer test documented acceptance scenarios using non-production data.
Launch approval	Owner approves actual contacts, policies, service claims, prices and operational capacity.
Post-launch	Monitor errors and notifications; review the first real bookings and reconcile first payments carefully.

Acceptance checklist

- ✓ No critical security issue, incorrect price calculation or false payment status remains open.
- ✓ Every advertised feature has a named owner and tested behavior.

✓ Keep a written issue list with severity, responsible person and target date.

20 / DELIVERY ROADMAP

Build in phases without losing control

Implement the platform in a sequence that delivers usable value early and avoids building a visually impressive but unreliable system.

PHASE 1

Foundation

Audit, UI kit, roles, records

PHASE 2

Operations

Requests, jobs, scheduling

PHASE 3

Commerce

Catalogue, inventory, quotes

PHASE 4

Finance

Invoices, payment verification

Interface modules

Phase 1 — Foundation

Audit existing code; approve visual system; configure business details; create data model, authentication, roles, backups and staging.

Phase 2 — Service operations

Build service pages, booking request flow, admin triage, scheduling, assignment, job status, technician checklist and customer updates.

Phase 3 — Retail and quotes

Add catalogue, product details, stock movements, supplier records, quotation builder, approval history and materials linked to jobs.

Phase 4 — Financial workflows

Invoice/receipt records, verified payment integration, partial payments, reconciliation and refund controls.

Phase 5 — Projects and insights

Contract/milestone workflows, maintenance reminders, reporting, customer portal enhancements and exports.

Phase 6 — Improve

Performance, SEO, accessibility, support process, staff training, analytics and planned maintenance.

Expected workflow

STAGE	BEHAVIOR / RULE
Estimate	Developer provides milestones, assumptions, exclusions and acceptance criteria before starting each phase.
Demonstrate	Review working staging features with real workflow examples at the end of each milestone.
Approve	Owner signs off before the next phase or production release.

Acceptance checklist

- ✓ Agree budget and timelines with the developer after source-code and integration audit.
- ✓ Do not promise a fixed implementation cost from this specification alone.

✓ Require code ownership, deployment access, documentation and handover arrangements in writing.

21 / REFERENCE MAPPING

How the supplied visual references translate to PhaseForge

Use the uploaded references as visual direction for layout and hierarchy, not as content to copy verbatim.

REFERENCE 1

Hero

Navy, gold, strong call to action

REFERENCE 2

Services

Six clear service cards

REFERENCE 3

Shop

Category-led product catalogue

REFERENCE 4

Booking/admin

Step flow + data dashboard

Interface modules

Electrical brand hero

Use the first reference's strong split hero, dark background and gold CTA, but PhaseForge logo, Ghana contacts and approved slogan.

Service cards

Use image-led cards with concise descriptions and Read More actions; use the six approved PhaseForge categories as the main taxonomy.

Product catalogue

Use the category sidebar/search and product grid pattern; ensure mobile filter controls and accurate Ghana stock/price labels.

Booking interface

Use numbered steps, a live request summary and clear review/submit action; do not display made-up appointment slots or foreign pricing.

Operations dashboard

Use KPI cards, work queues, charts and invoices panel; every metric must come from real records and respect roles.

Responsive behavior

Keep the visual hierarchy on mobile by stacking cards, collapsing navigation and simplifying wide tables.

Expected workflow

STAGE	BEHAVIOR / RULE
Design	Developer creates high-fidelity page mockups using PhaseForge colors and typography.
Review	Owner approves desktop and mobile views before full implementation.
Build	Implement reusable components and wire every visible action to real behavior.

Acceptance checklist

✓ Use licensed/owned imagery and the PhaseForge identity.
✓ Do not copy other brands' logos, text, pricing or country-specific claims.
✓ Use the provided images as design references only; final implementation must be original and business-accurate.

22 / DEVELOPER HANDOVER

Required handover package

The owner should be able to operate the business and change routine content without needing the developer for every update.

SOURCE CODE

Delivered

Repository and ownership

ACCESS

Documented

Owner-controlled accounts

TRAINING

Completed

Staff role walkthroughs

RECOVERY

Proven

Backup restore test

Interface modules

Technical handover

Repository, deployment procedure, architecture notes, database schema/migrations, environment variable names and integration inventory.

Operations handover

Admin guide for requests, jobs, stock, quotes, invoices, customers, reports and policy changes.

Account ownership

Owner-controlled hosting, domain, database, payment provider, email/WhatsApp provider and analytics accounts.

Training

Owner, dispatcher, technician, store and finance role walkthroughs using sample data.

Support plan

Defect warranty period, support response expectations, maintenance costs, update responsibilities and emergency contact.

Final acceptance

Signed checklist, known limitations, open issues and agreement on what is out of scope.

Expected workflow

STAGE	BEHAVIOR / RULE
Demonstrate	Developer shows each workflow live in staging/production as appropriate.
Transfer	Credentials are transferred securely; remove temporary developer access when no longer needed.
Verify	Owner tests routine tasks and confirms backups, alerts and access work.

Acceptance checklist

- ✓ Do not store shared admin passwords in the PDF or plain-text handover document.
- ✓ Owner retains control of domains, billing and production access.

✓ All known limitations and recurring third-party fees are documented.

APPENDIX A

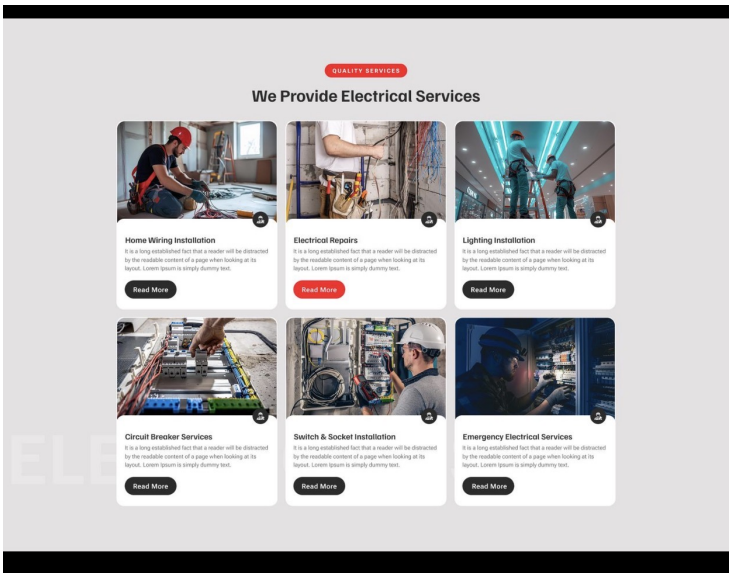
Visual references supplied by the owner

These images are treated as layout inspiration. Final copy, colors and details must be PhaseForge-specific and Ghana-appropriate.

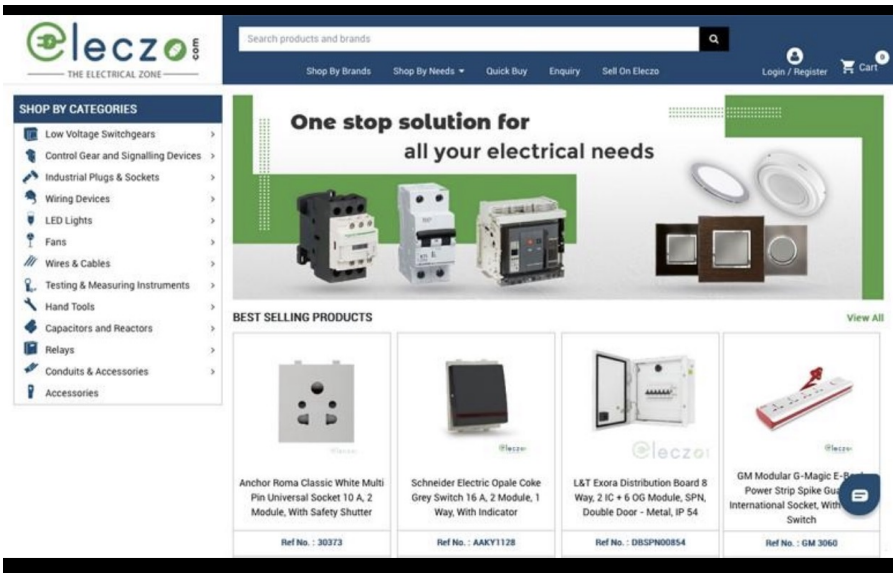
Reference 1



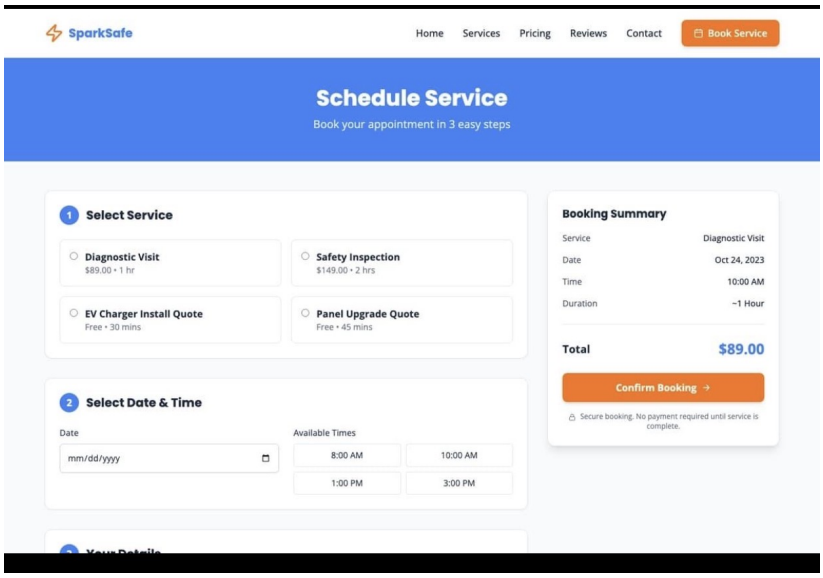
Reference 2



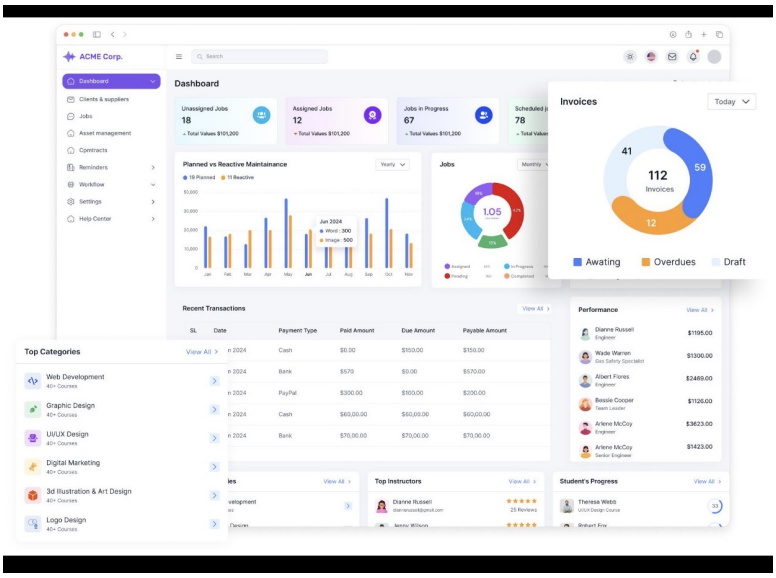
Reference 3



Reference 4



Reference 5


ACME Corp.
Dashboard
Unassigned Jobs
18
Assigned Jobs
12
Jobs In Progress
67
Invoices
Today
112
Invoices
41
59
12
Awating
Overdues
Draft
Planned vs Reactive Maintainance
Jobs
Recent Transactions
Performance
Top Categories
Web Development
Graphic Design
UI/UX Design
Digital Marketing
3d Illustration & Art Design
Logo Design
Top Instructors
Student's Progress

APPENDIX B

Launch sign-off checklist

Owner and developer should review and sign this before public launch.

■ Owner has approved all public text, contact numbers, service regions, hours, fees, credentials and warranty statements.
■ All six service categories have accurate scope, exclusions, safety notes and working request buttons.
■ Booking submissions save successfully, generate unique references and create staff work-queue records.
■ Admin roles and customer data isolation have been tested, including direct URL/API access.
■ Product stock, quote totals, invoice totals and payment statuses are accurate under success/failure/pending scenarios.
■ Photo uploads use private storage and tested file limits.
■ Notification failures are visible to staff; no false claim of delivery is shown.
■ Backups are configured and a restore test has been completed.
■ Privacy notice and service terms have been reviewed for Ghanaian requirements by appropriate professionals.
■ Owner controls domain, hosting, repository, database, payment and messaging accounts.
■ Mobile layout, accessibility, performance and all key links have been checked.
■ Open issues, support responsibilities, recurring fees and maintenance plan are documented.

Owner approval: _____ Date: _____

Developer approval: _____ Date: _____

Final note

This specification defines the intended experience and behavior. The developer must inspect the current source code and backend before estimating effort or promising a launch date. Features described here are requirements until implemented and tested; this PDF does not itself change the live website.

PHASEFORGE

COMPLETE DEVELOPER IMPLEMENTATION, TESTING & LIVE OPERATIONS SPECIFICATION

Addendum to the original 49-page Dashboard-Style Website Implementation Specification

Brand promise: POWER DONE RIGHT, every single time.

Prepared as a practical build-and-verification brief for the developer responsible for upgrading the existing PhaseForge website.

IMPORTANT DELIVERY STATUS

This document specifies what must be implemented and how it must be tested. It does not claim that the live website, database, payment gateway, notifications, or dashboard have already been changed. Actual implementation requires access to the website source repository, hosting project, environment configuration, database, and relevant provider accounts. Secrets must be shared through a secure secret manager—not pasted into this PDF or ordinary chat.

How to use this expanded specification

Keep the original 49-page specification as the visual and product brief. Append this implementation chapter to it. The original reference images establish the intended design language: a premium dark navy/gold electrical-services homepage, clear service cards, an electrical-products catalogue, a guided booking flow, and a data-driven operations dashboard.

The developer should treat this document as a delivery contract for scope, acceptance tests, security and operational readiness—not as a claim that these features already exist.

Non-negotiable delivery rule

A visible button is not a working feature. Every booking, quote, status change, stock adjustment, invoice, payment, notification and dashboard metric must be backed by validated server-side logic and persistent data. If an integration is not configured, show a clear unavailable state instead of displaying fake success.

Suggested delivery phases

Phase	Outcome	Exit condition
0. Access & audit	Source, hosting, database, integrations and existing flows understood	Written audit and backup/rollback plan approved
1. Foundation	Environments, database schema, authentication, roles, logs	Migrations and access-control tests pass
2. Public website	Homepage, service pages, contact and responsive design	Visual and content acceptance checks pass
3. Booking & jobs	Real requests, references, status workflow and assignments	End-to-end booking reaches staff dashboard
4. Products & stock	Catalogue, stock ledger, enquiries/orders	Stock changes are traceable and concurrency-safe
5. Quotes, invoices & payments	Approved quote flow, invoice records and verified payment callbacks	Amounts reconcile; duplicate callbacks are safe
6. Notifications	Email/WhatsApp/SMS where configured	Delivery status, retries and failure visibility work
7. Reporting & launch	Operational metrics, backup restore, monitoring and training	All launch gates signed off

1. Access, discovery and safe change management

1.1 Required access

• Read/write access to the source repository through named developer accounts with multi-factor authentication.

• Access to the Vercel project or current hosting provider, deployment settings, domains and DNS through least-privilege roles.

• Access to the database provider, migration history, backups and schema; do not expose production customer data in test environments.

• Access to transactional email, WhatsApp Business or SMS provider accounts only if the business chooses to use them.

• Payment provider merchant account access for the authorised owner or finance lead; use sandbox credentials during development.

• Existing booking forms, team login, file upload and GPS behaviour must be mapped before replacement.

• Named business decision-maker to approve service list, prices, service areas, emergency claims, warranty terms and customer policy wording.

1.2 Pre-change safety

• Record the current production commit, hosting configuration and database schema.

• Create a database backup and test that it can be restored to a separate environment.

• Capture screenshots and run smoke tests of existing pages and forms.

• Create staging and production environments with separate databases and separate provider keys.

• Use feature flags for new shop, payment and dashboard functions so unfinished work can be disabled safely.

• Document rollback steps, responsible person and expected rollback time before the first production release.

1.3 Audit deliverable

The developer must submit a short audit listing: existing routes/pages; current form endpoints; database tables and migrations; authentication and role model; image/file storage; notification pathways; deployment pipeline; known bugs; third-party services; and features that are only visual placeholders. No feature should be described as operational until its path from user action to persisted data and staff visibility has been demonstrated.

2. Recommended architecture and environments

2.1 Architecture principles

• Keep the current hosting platform if the repository and deployment setup are sound; do not rewrite solely for fashion.

• Use a managed relational database such as PostgreSQL for customers, bookings, jobs, products, stock movements, quotes, invoices and payments.

• Use server-side APIs or server actions for all privileged changes. The browser must never be trusted to set its own price, role, payment status or stock count.

• Store uploaded images and documents in private object storage with short-lived signed URLs where appropriate.

• Use an authentication provider or well-maintained framework, with server-enforced role permissions.

• Use a provider-supported payment integration for Ghana, selected after confirming merchant onboarding, fees, settlement and supported payment methods.

• Send notifications through an approved provider and record attempts, delivery results and failures.

• Centralise structured logs, error monitoring, uptime checks and audit events without logging passwords, access tokens or full payment credentials.

2.2 Environment separation

Environment	Purpose	Rules
Local development	Developer iteration	Synthetic data; no production secrets
Staging / QA	Acceptance tests and stakeholder review	Separate database and sandbox payment keys; safe test phone/email recipients
Production	Real customers and transactions	Restricted access; backups; monitored releases; approved migrations only

2.3 Proposed system map

Customer browser/mobile → public website → validated server API → database/storage → job queue or reliable notification service → email/WhatsApp/payment provider. Staff dashboard → authenticated server API → same business records. Payment provider callbacks must be authenticated and reconciled server-side before an invoice is marked paid.

2.4 Data integrity rules

• Store money as integer minor units or a documented decimal type; never use floating-point arithmetic for totals.

• Use a single documented timezone strategy and display Ghana local time consistently to customers and staff.

• Every important record has a stable ID, created/updated timestamps and an appropriate status.

• Use database transactions for operations that must succeed together, such as creating an order and reserving stock.

• Use idempotency keys or equivalent duplicate protection for payment callbacks and retryable actions.

• Use soft deletion or archival for financial and operational records where an audit trail is required.

3. Core database model

The exact schema depends on the existing application. The following logical entities are the minimum recommended model; the developer should map or migrate existing tables rather than create duplicate sources of truth.

Entity	Key fields / relationships	Critical rule
User / StaffProfile	user_id, name, email/phone, role, active	Role checked on every protected server action
Customer	customer_id, name, contacts, preferred channel	Collect only necessary personal data
ServiceCategory / Service	name, description, active, qualification notes	Only advertise services actually offered
BookingRequest	reference, customer, service, location, urgency, preferred window, description, attachments, status	Request is not a confirmed appointment until accepted
Appointment	booking_id, assigned team, start/end, confirmation state	Prevent conflicting confirmed appointments
Job	job_id, booking_id, scope, assigned technician, status, safety notes	Every status transition is validated
JobEvent / AuditEvent	actor, entity, action, timestamp, before/after summary	Append-only where practical
Product / SKU	SKU, name, category, unit, price, active, reorder level	Prices and availability are server-controlled
StockMovement	product, quantity delta, reason, source record, actor	Never silently overwrite stock balance
Quote / QuoteLine	version, scope, quantities, labour, materials, taxes/fees if applicable, expiry, approval	Approved quote version is immutable; revisions create a new version
Invoice / InvoiceLine	number, customer, source job/order, totals, due date, status	Totals recomputed server-side
Payment / PaymentEvent	provider reference, amount, currency, status, timestamps	Paid status only after verified provider evidence
Notification / DeliveryAttempt	recipient, channel, template, attempt, status, provider ID	Failures visible to staff and retried safely
MaintenanceContract	site, covered assets, schedule, exclusions, renewal	Contract dates and scope explicit

3.1 Status dictionaries

Booking: received → under review → needs information / quote required → accepted → scheduled → completed / cancelled / declined. **Job:** awaiting assignment → assigned → en route (optional) → on site → in progress → awaiting parts / customer approval → work completed → quality check → closed. Use a separate status for customer cancellation and staff cancellation.

Quote: draft → sent → viewed (if tracked) → approved / declined / expired / revised. **Invoice:** draft → issued → partially paid → paid / overdue / void. **Payment:** initiated → pending → successful / failed / cancelled / refunded. These are examples; developer must document allowed transitions and prevent illegal jumps.

3.2 Reference generation

Generate a non-sequential public reference such as PF-26-XXXXXX using a collision-safe server method. Do not expose database primary keys. Reference lookup must require additional verification or a secure customer account; a reference alone should not reveal sensitive job information.

4. Authentication, roles and permissions

Role	Allowed access (illustrative)	Must not be allowed by default
Owner / Super Admin	All configuration, staff roles, reports, records and audit trail	Share login; bypass audit logging
Operations Manager	Bookings, jobs, assignments, service schedule, customer contact	Change payment provider credentials or grant self super-admin
Technician	Assigned jobs, approved scope, site notes, job photos and completion checklist	See unrelated customer records or edit invoice/payment records
Sales / Storekeeper	Product catalogue, enquiries, stock receipts and adjustments	Alter completed job notes or mark payment successful without verification
Finance	Quotes/invoices, payment reconciliation, refunds subject to approval, financial reports	Edit technical test results or erase audit trail
Customer	Own bookings, quotes, invoices and permitted job updates	View another customer's records
Website visitor	Public pages, public products and booking request form	Access any staff API or private file

4.1 Permission implementation

• Enforce access checks on the server for every read and write, not only by hiding dashboard menu items.

• Default to deny when a role or permission is missing.

• Require multi-factor authentication for privileged staff where supported.

• Deactivate former staff accounts promptly and revoke sessions.

• Rate-limit login, password reset, public reference lookup and form submission endpoints.

• Record sensitive actions: role changes, refunds, stock adjustments, quote approvals, invoice voids, job closure and customer data exports.

• Test direct API calls with a lower-privileged account; UI hiding alone is not an access-control test.

4.2 Account recovery

Password reset tokens must be short-lived, single-use and never logged. The reset response should not reveal whether an email address is registered. Staff account recovery should include an owner-controlled process. Do not store passwords in plain text or build custom cryptography.

5. Public website: visual and functional acceptance

5.1 Design system

• Use PhaseForge branding and the slogan “POWER DONE RIGHT, every single time.”

• Use a consistent dark navy, gold, white and neutral-grey palette inspired by the supplied reference; maintain accessible contrast.

• Create reusable components for page headers, buttons, service cards, product cards, form fields, status badges, dashboard KPI cards, tables, filters, empty states, loading states and error messages.

• Keep typography, spacing, border radius, icon weight and button hierarchy consistent across all sections.

• Use real PhaseForge photography where available; do not imply staff, certification, equipment or customer endorsements that are not genuine.

• Do not copy the UK brand's NICEIC claim, UK pricing, UK phone numbers or UK-specific compliance wording from the inspiration image.

• Make mobile layouts first-class: navigation, cards, forms, tables and dashboard filters must remain usable on small screens.

5.2 Homepage

• Hero: clear local value proposition, service area and two actions: Request a Quote and Explore Services.

• Service shortcuts for Home Electrical Services; Commercial & Industrial; Solar & Backup Power; Testing & Safety; Emergency Assistance; Projects & Contracts.

• Product spotlight for cables, breakers, switches/sockets, conduits, LED lighting and accessories, only where actual stock or sourcing is available.

• Trust strip with factual statements only. Remove or verify claims such as job totals, years of experience, 24/7 response, guarantees, licensing, same-day callouts and no-repeat-visit promises.

• Show contact routes and hours that the business can actually support.

• Include clear footer links to contact, privacy notice, customer terms, cancellation policy and service-area coverage.

5.3 Service detail page acceptance

Every service page should explain: what is included; typical reasons to request it; information needed to quote; what is excluded; whether an inspection is required; safety precautions; whether materials can be supplied; how pricing is determined; expected next steps; and how to request a visit. Do not publish technical guarantees or compliance certification claims without approval.

5.4 Content localisation

Use Ghanaian customer language, GHS currency and Ghana local time. Mention Kasoa, Tema, Kpone, East Legon and wider Greater Accra only to the extent coverage is confirmed. For other regions, present an enquiry route rather than promising nationwide response unless operations support it.

6. Booking workflow: build and test

6.1 Customer journey

Step	Customer action	System response
1. Select request type	Service visit, quote request or emergency enquiry	Load relevant service-specific questions
2. Describe work	Select service, property type, symptoms and urgency	Validate required fields and show safety warning where relevant
3. Add location	Town/area, address, optional map pin and access notes	Check service-area coverage; allow enquiry where outside area
4. Add evidence	Optional photos/documents with clear size/type limits	Validate on server, store privately and scan where available
5. Choose availability	Preferred date/time windows and contact preference	Show as preference only unless real capacity is connected
6. Review & consent	Confirm details, privacy notice and terms	Display what happens next and any disclosed call-out/assessment fee
7. Submit	Send request once	Persist record, issue reference and show success only after save
8. Staff processing	Review, request info, quote, accept or decline	Record actor/time; send notifications; update customer-visible state
9. Confirm appointment	Customer accepts agreed time and disclosed fees	Appointment becomes confirmed and calendar capacity is reserved

6.2 Validation requirements

• Validate required name, a reachable phone/WhatsApp contact, service, description, location and consent as applicable.

• Normalise Ghana phone numbers while preserving the customer's entered value for display.

• Validate uploaded file type by content where possible, not just filename; enforce size and count limits server-side.

• Use anti-spam controls and rate limits; avoid CAPTCHA friction unless needed.

• Prevent double submission using an idempotency token or equivalent request protection.

• Never show “booking confirmed” merely because a form was submitted. The initial state is “request received” until staff or scheduling logic confirms it.

• GPS/location access must be optional, permission-based and paired with manual address entry.

• Customer must be able to request a quote without committing to work or payment unless a disclosed booking fee applies.

6.3 Booking tests

Test	Expected result
Submit valid ordinary request	One persisted request, unique reference, confirmation screen
Submit without required field	Inline validation; no database record
Double-tap submit / refresh	No duplicate booking
Upload oversized or disguised executable	Rejected server-side with helpful message
Deny GPS permission	Manual location entry remains usable
Out-of-area request	Accepted as enquiry or politely declined according to policy; no false coverage promise
Staff accepts a date/time	Appointment confirmed; customer receives correct details
Staff changes or cancels appointment	History recorded and customer notified
Notification provider fails	Request still saved; failure logged and retry/manual follow-up visible

7. Job operations dashboard

7.1 Dashboard layout

Use a compact dashboard like the supplied management reference: top navigation/search; left navigation; KPI cards; charts or trend panels; recent jobs; pending approvals; invoice/payment summary; inventory alerts; and an activity feed. Avoid decorative charts with invented data.

7.2 Required navigation

- Overview
- Bookings & Schedule
- Jobs & Dispatch
- Customers
- Quotes
- Invoices & Payments
- Products & Inventory
- Suppliers & Purchase Receipts (if used)
- Maintenance Contracts
- Technicians & Availability
- Reports
- Notifications / Failed Deliveries
- Audit Log
- Settings & Permissions

7.3 KPI definitions

Metric	Definition	Controls
New requests	Requests received in selected period	Exclude spam/deleted records by documented rule
Awaiting review	Requests not yet triaged	Click opens filtered queue
Scheduled jobs	Confirmed appointments in selected period	Do not count unconfirmed preferences
Jobs in progress	Jobs currently in valid active statuses	Exclude closed/cancelled jobs
Quotes awaiting response	Sent quotes not approved/declined/expired	Show age and follow-up action
Unpaid invoices	Issued invoice balance greater than zero	Reconcile partial payments and credits
Low-stock products	Active SKUs at/below reorder level	Exclude inactive/discontinued SKUs
Collected payments	Verified successful payments in selected period	Use provider-confirmed records and refund adjustments
Overdue jobs	Jobs past due date or agreed service target	Display target definition and exclusions

7.4 Dashboard behaviours

- All KPI cards must link to the underlying filtered records.
- Every chart has a date range, clear labels, meaningful empty state and accessible table alternative.
- Date range, status and assigned technician filters must apply consistently to all relevant data.

- Do not cache financial or job counts so long that staff see misleading states; document cache policy.
- Loading, empty, partial-failure and permission-denied states must be designed, not left blank.
- Use pagination and server-side filtering for large job/product lists.
- Provide CSV export only to authorised roles and log sensitive exports.

8. Job lifecycle and technician workflow

8.1 Required operational stages

• Intake: verify customer contact, service, location, urgency, hazards and scope.

• Triage: decide whether remote clarification, site assessment, quote or emergency escalation is appropriate.

• Estimate: separate labour, materials, travel/call-out, third-party charges and exclusions where applicable.

• Approval: capture customer acceptance of a specific quote version and any deposit or payment terms.

• Assignment: choose qualified and available staff; provide only information needed for the job.

• Pre-visit: confirm access, parking/site contact, power isolation requirements and customer preparation.

• Arrival: record check-in and any change in scope or safety conditions.

• Work: record diagnosis, approved work, materials used, tests performed, photos where appropriate and any deviations.

• Change control: stop and seek approval before chargeable work outside the agreed scope, except immediate actions necessary to protect life/property under applicable law and professional judgment.

• Completion: document work done, test results, limitations, customer handover and remaining recommendations.

• Quality check: review checklist, materials, photos and customer explanation.

• Closure: issue invoice/receipt as applicable, update inventory, set warranty/workmanship record and invite feedback without pressure.

8.2 Technician job view

On a phone, the technician should see today's assigned jobs, map/address, contact method, service scope, approved quote, safety notes, required materials, check-in/out, job checklist, parts used, customer approval request, photo upload and completion summary. The interface must not expose unrelated customer jobs.

8.3 Safety-critical controls

The system should support stop-work and escalation notes. A technician must be able to record unsafe access, exposed live parts, suspected upstream/network faults, water ingress, fire damage, missing isolation, inadequate earthing or other conditions requiring work to pause. The software must not encourage unqualified persons to work on live equipment.

8.4 Job record audit

Record who changed status, assignment, quote linkage, scope, schedule and closure, including timestamps. Corrections should preserve the prior value or history. Do not allow a user to erase evidence of completed work or safety observations.

9. Product catalogue and inventory

9.1 Customer-facing product catalogue

• Categories: wires & cables; circuit breakers and protection; distribution boards; switches, sockets and plugs; conduits and fittings; LED lighting; electrical tools and testing equipment; solar/inverter/battery systems; generators and accessories; consumables.

• Each listing should include product name, brand/model when verified, key specifications, unit, availability state, price or enquiry-only label, warranty information supplied by the manufacturer and product image rights.

• Clearly distinguish in-stock, low-stock, available on order and discontinued items.

• Do not publish a price until the business has approved it and established how long it remains valid.

• Provide enquiry/cart/quote request based on business capability; do not imply online checkout works until payment, order creation and stock rules are implemented.

9.2 Inventory rules

• Every SKU has a unique identifier, unit of measure, reorder level and active/inactive state.

• Stock receipts, sales, returns, damaged items, transfers and manual adjustments create stock movement records with reason and actor.

• Stock on hand is calculated from a reliable ledger or updated transactionally with a ledger; do not silently overwrite quantity.

• Only authorised store roles can adjust stock; large adjustments require a second approval if the owner chooses that control.

• Prevent negative stock unless explicitly allowed for backorders and visibly labelled.

• Concurrent orders must not oversell stock; reserve or decrement stock within a database transaction.

• Track cost price separately from retail price and restrict cost/margin visibility to authorised roles.

• Low-stock alerts must respect active status and reorder level, and should avoid repeated duplicate alerts.

9.3 Inventory tests

Scenario	Pass condition
Receive 10 units	Stock and ledger reflect +10 with receipt reference
Sell 3 units	Stock reduces by 3 only once
Two customers order last unit simultaneously	Only one reservation succeeds or backorder policy is applied
Return 1 unit as damaged	Stock disposition is explicit; sellable quantity not incorrectly increased
Unauthorised technician attempts adjustment	Server denies and logs attempt
Product reaches reorder level	Alert appears in dashboard and can be acknowledged
Deactivate a SKU	Existing historical transactions remain; new sale blocked

10. Quotations, invoices, payments and reconciliation

10.1 Quote content

• Business identity and verified contact details.

• Quote number, issue date, expiry date, customer/site details and a plain-language scope.

• Labour, materials, quantities, unit prices, transport/call-out, third-party costs and applicable taxes/charges shown transparently where relevant.

• Explicit assumptions, exclusions, access requirements, provisional items, expected schedule and quote validity.

• Deposit/milestone/final payment terms, cancellation/variation rules and warranty/workmanship terms.

• Acceptance method that identifies the approved version and timestamp.

10.2 Invoice controls

• Invoice number is unique and generated server-side.

• Totals are recalculated from stored lines on the server; client-supplied totals are ignored.

• Partial payments and credit notes are represented accurately.

• Voiding or refunding requires permission and a reason; preserve the original record.

• Receipt is issued only for a recorded payment; a payment screenshot alone is not proof of settlement.

• Currency and any tax treatment must be confirmed with the business's accountant or Ghana tax adviser.

10.3 Payment integration requirements

Select a Ghana-supported provider only after verifying the provider's current merchant requirements, supported payment methods, settlement process, transaction fees, refund process and webhook documentation. Use sandbox mode for tests. Do not hardcode provider-specific assumptions into the business workflow.

• Create a pending payment record before redirecting or initiating a payment.

• Verify provider callback signatures or authenticate status checks according to official provider guidance.

• Match provider reference, expected amount, currency and intended invoice/order.

• Mark paid only after authoritative server-side verification; never trust a browser redirect alone.

• Make callback processing idempotent so duplicate events cannot duplicate payment or order records.

• Handle pending, failed, cancelled, expired, refunded and disputed states.

• Reconcile daily provider settlements against internal payment records and bank/mobile-money statements.

• Keep secrets in environment variables/secret storage and rotate them if exposed.

10.4 Financial acceptance tests

Test	Expected result
Tamper with browser total	Server uses stored prices and rejects altered total
Fake success redirect	Invoice remains unpaid until verified
Duplicate provider callback	One successful payment record; no duplicate fulfilment
Wrong amount or currency	Flagged for review; not auto-marked paid
Partial payment	Outstanding balance recalculated correctly
Refund	Permission and reason required; original payment remains auditable
Provider unavailable	Customer sees pending/failure guidance; no false payment success

11. Notifications and communications

11.1 Event-driven messages

Event	Customer message	Internal action
Request received	Reference and next steps; not yet confirmed appointment	Create triage task
Need more information	Specific questions and secure reply route	Mark awaiting customer
Quote sent	Scope, total, expiry and approval route	Create follow-up reminder
Quote revised	Explain version changed and prior approval status	Keep version history
Appointment confirmed	Date/time window, location, contact and preparation	Reserve schedule
Appointment changed/cancelled	Reason and revised plan where known	Update calendar and audit event
Technician assigned / on the way	Only if location/dispatch data is reliable	Update job event
Job completed	Summary, invoice/payment next step and support contact	Trigger QA/closure workflow
Payment verified	Receipt and invoice balance	Reconcile payment record
Notification failed	No customer-facing false claim	Queue retry and show failure in admin

11.2 Message quality

• Use approved templates and the customer's selected contact channel where possible.

• Keep messages concise, factual and consistent with current status.

• Do not send sensitive photos, payment data or detailed personal information to a shared/unverified number.

• Provide a clear route to contact the company and correct wrong contact details.

• Respect opt-outs and applicable Ghanaian privacy/communications requirements.

• Store provider message IDs and delivery state without retaining unnecessary message content.

11.3 Retry and failure handling

Use bounded retries with backoff for temporary failures, plus a visible dead-letter/manual follow-up queue. A notification failure must not delete or roll back a valid booking. Alert an authorised staff member when a high-priority emergency request cannot be delivered to the expected team.

12. Privacy, security and Ghana-specific business controls

12.1 Customer data

• Publish a privacy notice explaining data collected, purpose, recipients/processors, retention, customer rights and contact route.

• Collect only details needed to assess and deliver the service; make GPS and photo upload optional unless genuinely necessary.

• Restrict access to customer contact information, property addresses, photos, job history and invoices by role.

• Define retention and deletion rules with legal advice, accounting obligations and legitimate operational needs in mind.

• Use HTTPS, secure cookies, CSRF protection where applicable, output escaping, parameterised queries and secure upload handling.

• Maintain dependency updates, vulnerability reporting, access reviews and an incident-response process.

• Confirm applicable Ghana Data Protection Act and Data Protection Commission obligations with a qualified adviser before launch.

12.2 Electrical business and compliance claims

The website and policy must be reviewed against applicable Ghanaian electrical safety requirements and any relevant Energy Commission, Electricity Company of Ghana, Ghana Standards Authority, local authority, insurance and tax requirements that apply to the particular work. The developer must not invent a licence, certification, insurance status or compliance guarantee. The business owner should confirm qualifications and permitted scope for each advertised service.

12.3 Consumer terms

Have a Ghana-qualified lawyer or appropriate adviser review the final customer terms, cancellation policy, deposit policy, workmanship warranty, limitation language, dispute process, privacy notice and any tax or statutory wording. Terms must not attempt to remove non-excludable statutory rights or excuse negligent or unsafe work.

12.4 Emergency messaging

For immediate danger, fire, electric shock or exposed live equipment, the website should instruct customers to keep people away and contact the appropriate emergency/utility service as needed. Do not instruct customers to touch, open or repair electrical equipment. Emergency callouts must be advertised only when staffing and response coverage can genuinely support them.

13. Testing strategy and evidence

13.1 Test layers

• Unit tests: validation, pricing calculations, status transition rules, permissions and reference generation.

• Integration tests: database writes, file storage, notification provider, payment sandbox and webhook processing.

• End-to-end tests: customer submits request → staff sees it → quote issued → approval captured → job scheduled → job completed → invoice/payment reconciled.

• Security tests: role bypass, direct API calls, upload abuse, rate limits, session handling and secret exposure.

• Accessibility checks: keyboard operation, focus visibility, labels, contrast, error association and screen-reader structure.

• Responsive visual checks: common mobile, tablet and desktop widths; long names; large tables; slow network; empty/error states.

• Recovery tests: restore backup to a clean environment, retry failed notifications, recover interrupted payment status and roll back a bad release.

13.2 Browser/device matrix

Target	Minimum check
iPhone Safari	Navigation, booking, file picker, tap targets, call/WhatsApp links
Android Chrome	Forms, uploads, responsive cards, GPS denied/allowed
Desktop Chrome	Dashboard tables, filters, modals, keyboard flow
Desktop Safari / Firefox / Edge	Core page rendering and booking completion
Slow or unstable connection	Loading states, safe retry, no duplicate form/payment actions

13.3 Defect severity

Severity	Example	Release rule
P0 Critical	Data exposure, unauthorised admin access, wrong payment state, lost bookings	Block release; fix and retest
P1 High	Booking cannot be submitted, stock oversells, invoice totals wrong, emergency flow fails	Block release unless formally mitigated and approved
P2 Medium	Filter broken, notification delay, layout issue affecting some devices	Fix before launch where core workflow affected; otherwise documented owner/date
P3 Low	Minor visual inconsistency or non-blocking copy issue	Can be scheduled after launch if approved

13.4 Evidence package

• Test plan with IDs, steps, expected result, actual result, pass/fail and tester/date.

• Sanitised screenshots or screen recordings of key workflows.

• Automated test output and known limitations.

• Payment sandbox transaction references (never credentials or card secrets).

• Backup and restore evidence.

• Security/permission test evidence.

• List of unresolved defects and explicit business acceptance for any non-critical items.

14. End-to-end acceptance scenarios

ID / flow	Action	Acceptance criteria
E2E-01: Customer booking	Submit a residential fault request with valid contact and location.	One request and reference persist; appears in staff queue; confirmation says request received, not confirmed appointment.
E2E-02: Staff triage	Operations staff requests additional information.	Status and question are recorded; customer notification is attempted; failure is visible.
E2E-03: Quote approval	Create and send a versioned quote; customer approves that version.	Approval timestamp and quote version persist; job can proceed using the approved scope.
E2E-04: Schedule & assign	Assign a qualified available technician and confirm a date/time.	Conflicts are prevented or flagged; customer and technician receive correct details.
E2E-05: Scope variation	Technician finds additional work outside quote.	System records change request; customer approval required before chargeable additional work.
E2E-06: Completion	Technician enters work summary, materials, test checklist and handover.	Required fields are checked; job moves through quality check and closure.
E2E-07: Invoice/payment	Issue invoice, make sandbox payment, process callback.	Verified amount is matched; invoice balance and dashboard update once.
E2E-08: Stock usage	Record material used on job.	Stock ledger and job record match; low-stock alert triggers where applicable.
E2E-09: Role restriction	Technician tries to open another customer's job or edit invoice.	Server denies action and logs security-relevant event.
E2E-10: Failure recovery	Email provider fails during request submission.	Request persists; failure is visible and retry/manual follow-up is possible.
E2E-11: Cancellation	Customer asks to cancel under published terms.	Staff records outcome, any applicable fee/refund decision and sends confirmation.
E2E-12: Backup restore	Restore backup into isolated environment.	Database integrity checks pass; recovery steps documented.

15. Performance, reliability and monitoring

15.1 Performance targets

Agree targets with the developer after baseline measurement. Suggested starting goals: public pages should feel responsive on a typical Ghana mobile connection; avoid oversized hero images; load only needed scripts; paginate large tables; and keep form submission feedback immediate. Treat these as goals to measure, not guarantees.

• Compress and correctly size images; use responsive formats and lazy-load below-the-fold media.

• Set reasonable request timeouts and handle upstream provider errors gracefully.

• Add database indexes for common filters: status, created date, assigned technician, customer and SKU where appropriate.

• Monitor page errors, API error rate, latency, failed background tasks, database health and payment webhook failures.

• Create alerts for booking submission failures, payment reconciliation mismatches, failed emergency notifications and backup failures.

• Provide an incident log with time, impact, owner, mitigation and resolution.

15.2 Operational dashboard status

Any live system-health indicator must be connected to real checks. If the database or notification provider has not been checked, show “Status unknown” rather than “All good.” Never display simulated electrical panel, generator or surge-protection readings as if PhaseForge monitors a customer's real installation.

15.3 Backups

• Define backup frequency and recovery point objective with the owner.

• Encrypt backups and restrict access.

• Test restore at a documented interval and after major schema changes.

• Include database, essential configuration and uploaded-file recovery in the plan.

• Keep provider secrets out of database exports and PDFs.

• Document who can initiate restore and how customers/jobs created after the restore point are reconciled.

16. Deployment, release and rollback

16.1 Release checklist

• All approved code reviewed and merged; no unreviewed direct production edits.

• Database migration tested against a production-like copy and has a rollback or forward-fix plan.

• Environment variables present in correct environment; no secret values in source code or client bundle.

• Staging acceptance tests signed off by business owner.

• Payment and notification integrations switched from sandbox only after owner approval.

• Backups verified; monitoring and error reporting enabled.

• Customer-facing copy, prices, coverage and policies approved by the owner.

• Support contact, incident owner and rollback decision-maker identified.

• Release deployed in a controlled window and smoke-tested immediately.

16.2 Post-release smoke test

Check	Pass condition
Homepage and service pages	Load without critical visual or console errors
Booking submission	Test request saved once and visible to authorised staff
Staff login/roles	Correct access; unauthorised pages and APIs blocked
Product listing	Correct price and availability from database
Quote/invoice	Totals and status accurate
Notification	Test message sent to approved test recipient or failure visibly logged
Payment	Sandbox/approved test flow reconciles correctly
Mobile experience	Core journey usable on phone
Monitoring/backup	Health signals and backup schedule active

16.3 Rollback triggers

Rollback or disable the affected feature if a release exposes customer data, loses bookings, miscalculates money, accepts unauthorised actions, incorrectly marks payments paid, or causes widespread failure. Preserve logs and evidence, notify the designated owner, restore service safely, then investigate before re-enabling.

17. Owner responsibilities and approvals

Decision	Owner must confirm before launch
Service scope	Which residential, commercial, industrial, solar, generator and testing services are genuinely offered
Qualifications	Which services require particular training, licences, authorisation or specialist subcontractors
Coverage	Actual towns/regions served and travel/call-out arrangements
Emergency promise	Whether 24/7 availability is staffed and what response time can realistically be promised
Pricing	Call-out fee, assessment fee, quote validity, deposits, travel, taxes/charges and payment terms
Materials	Stocked items, sourcing lead times, substitutions, returns and manufacturer warranty handling
Workmanship warranty	Duration, coverage, exclusions and claim procedure
Cancellation/refund	Notice periods, deposits, completed work, special-order goods and refund authority
Data privacy	Privacy notice, retention, access and incident contact
Legal/compliance	Qualified review of customer terms and relevant Ghana obligations
Payment provider	Merchant onboarding, methods, settlement, refunds and reconciliation owner
Support	Who monitors bookings, failed notifications, jobs, inventory and payment exceptions

Approval rule

The developer implements the approved business rules; the owner approves commercial claims and customer-facing terms. The developer must not guess these decisions or quietly publish placeholder values.

18. Indicative implementation schedule

This is a planning sequence, not a guaranteed estimate. Timing depends on code quality, integrations, existing backend and developer capacity.

Workstream	Typical effort range	Dependency / exit
Audit, architecture and backups	2–5 working days	Access granted; current workflows mapped
Design system and responsive public pages	4–10 working days	Brand and content approved
Database/auth/roles and audit trail	5–12 working days	Schema and role model approved
Booking, staff triage and job workflow	6–15 working days	Real notification requirements confirmed
Product catalogue and stock ledger	5–12 working days	SKU, pricing and stock policy ready
Quotes, invoices and payment integration	7–18 working days	Provider account, sandbox and finance rules ready
Notifications and reporting	4–10 working days	Templates and recipients approved
QA, security checks, UAT and launch	5–12 working days	All workflows integrated

Several tasks can overlap, but payment integration, stock concurrency, security and real end-to-end testing should not be skipped to meet a deadline.

19. Developer handover package

• Source code in the agreed repository with meaningful commits and setup instructions.

• Architecture overview and environment variable names (never secret values).

• Database schema/migrations, backup and restore instructions.

• API route/action documentation and authentication/permission matrix.

• Status transition rules and business logic notes.

• Test suite and reproducible test instructions.

• Deployment, rollback and incident-response runbook.

• Payment provider setup notes, webhook verification and reconciliation procedure.

• Notification templates, provider status handling and retry policy.

• Admin/user guide for bookings, jobs, quotes, invoices, products, stock and reports.

• Known limitations and outstanding defects, each with severity and owner.

• Training session and sign-off by the PhaseForge owner.

Definition of done

A feature is done only when it is implemented in the agreed environment, backed by persistent data where needed, protected by server-side permissions, tested for success and failure paths, usable on mobile, monitored where operationally important, documented, and accepted by the business owner.

What must never be represented as completed without evidence

• Live payment collection or settlement

• WhatsApp/SMS/email delivery guarantees

• Live stock counts or availability

• Real-time dashboard health readings

• Confirmed appointment capacity

• Electrical certification, licence or insurance status

• Backups that have never been restored in a test

• Security claims that have not been tested

20. Final launch sign-off sheet

Sign-off area	Pass / fail	Evidence / owner / date
Visual design matches approved PhaseForge reference and works on mobile		
Service content, contacts, coverage, hours and claims approved		
Booking creates one real record and is visible to staff		
Staff roles and API permissions tested		
Job assignment, status history and completion record tested		
Product catalogue and stock ledger tested		
Quotes, invoices and totals verified		
Payment verification and reconciliation tested in approved environment		
Notifications and failure recovery tested		
Privacy, customer terms and legal review completed		
Backup restore and rollback demonstrated		
Monitoring, support ownership and training completed		
Known defects reviewed and accepted or resolved		

Approval

Business owner: _____ Date: _____

Lead developer: _____ Date: _____

QA / tester: _____ Date: _____

Release version / commit: _____ Production release date: _____

Final note: the original PDF and this addendum together define the intended website experience and the practical controls needed to make it operational. The real live implementation still requires source-code and hosting access, actual configuration, deployment, and the passing of the tests described here.